# Linear Algebra Notes

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## 1 Vector Spaces

Definition: A vector space $V$ is a set equipped with addition and scalar multiplication…

### 1.1 Subspaces

If $W \subseteq V$ and $W$ is also a vector space, then $W$ is called a subspace of $V$.

## 2 Eigenvalues and Eigenvectors

For a matrix $A \in \mathbb{R}^{n \times n}$, if there exists a non-zero vector $v$ and a scalar $\lambda$ such that:

$$Av = \lambda v$$

then $\lambda$ is called an eigenvalue, and $v$ is the corresponding eigenvector.

### 2.1 Characteristic Equation

Eigenvalues can be found by solving the characteristic equation:

$$\det(A - \lambda I) = 0$$